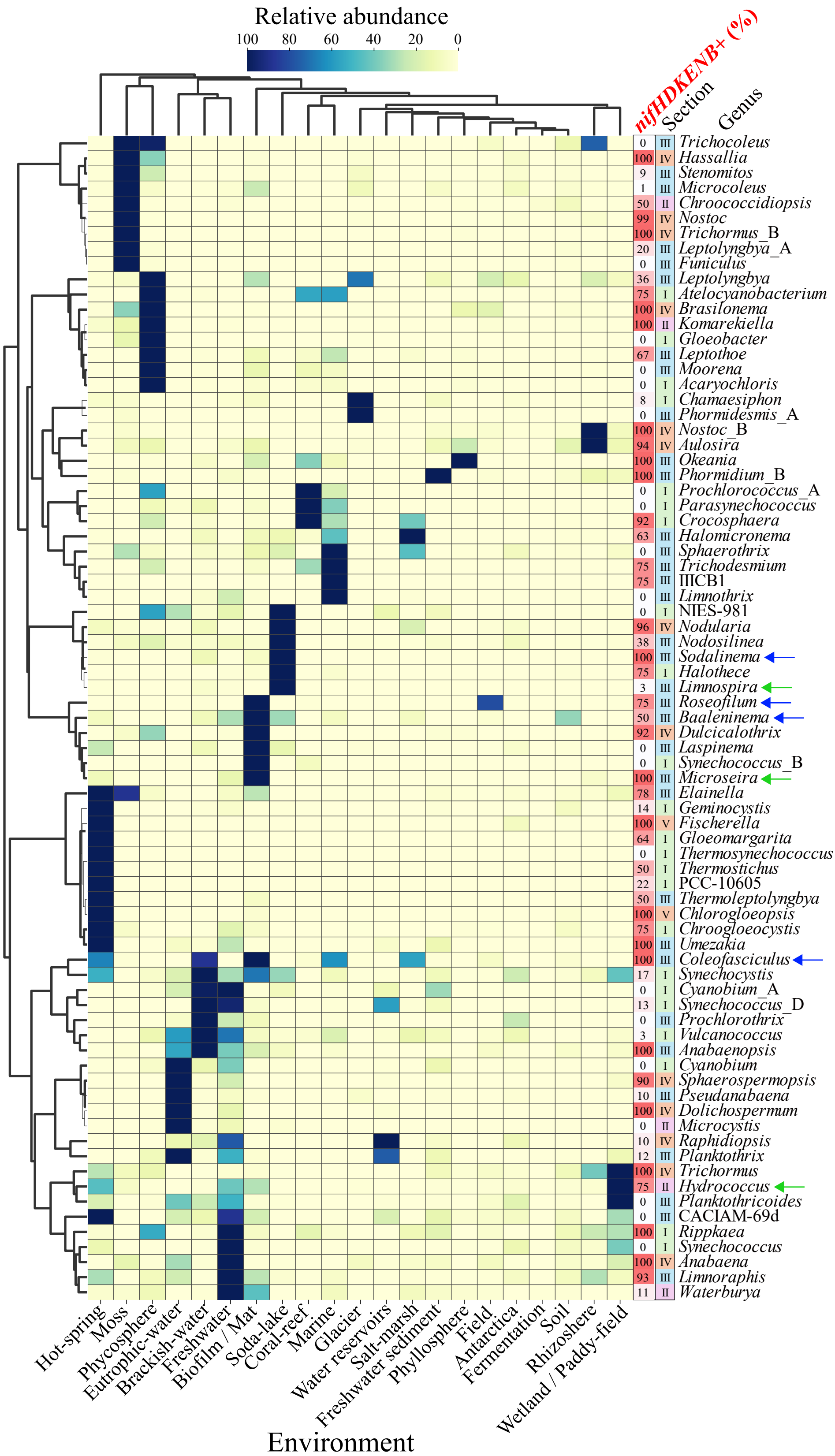

Figure 6. Metagenomic profiling of cyanobacteria at the genus level across diverse environments. Abundance profiles of 77 major cyanobacterial genera were investigated based on counts of 1.26 million publicly available metagenomic sequencing from the NCBI Sequence Read Archives. For each SRA/ERA/DRA dataset, 300,000 reads were subsampled and analyzed using the Kraken2 taxonomic profiler against a custom database of 2,716 GTDB R226 cyanobacterial representative genomes. Relative abundance below 0.1% were excluded, and average abundances were row normalized within each genus (scaled 0–100) to highlight distribution patterns across diverse environmental niches. Hierarchical clustering was performed using Ward's method based on Euclidean distances. Sections I–V and FNF (%) per genus (*nifHDKENB*+) are displayed alongside the genus names. Four genera containing strains with the ancient *nif* (Group II) genes are shown with blue arrows and three genera containing strains with both canonical (Group I) and ancient (Group II) *nif* genes are shown with green arrows.